Vraj Patel

Bengaluru, Karnataka - 560029
+91 8320884603 vrajpat3ll@gmail.com linkedin.com/in/vrajpat3ll github.com/vrajpat3ll

Education

Indian Institute of Technology, Jammu **Nov 2022 – May 2026**
Bachelor of Technology in Computer Science and Engineering | CGPA: 8.19 / 10 *Jammu, J&K*

Experience

Accenture, India **Jul 2026 – Present**
AI Native Software Engineering Analyst *Bangalore, Karnataka*

- Developing full-stack applications using C#.NET and ASP.NET, owning features across application development and AI engineering, applying solid software engineering practices.

Accenture, India **May 2025 – Jul 2025**
Advanced Application Engineering Analyst Intern *Bangalore, Karnataka*

- Automated cloud capacity monitoring and utilization reporting across 10+ Azure subscriptions, improving visibility into infrastructure usage for operational planning.
- Built REST API-driven analytics and event-based reporting pipelines with real-time dashboards to improve observability of cloud resource consumption.
- Developed an AI-assisted infrastructure provisioning workflow combining Terraform and LLM APIs to automate deployment decisions and workflows.

Projects

Orion – Terminal-Based Coding Agent | *Python, OpenAI, Docker, GitHub Actions, Tavily* **Link**

- Designed and built an autonomous coding agent that executes multi-step software development workflows through LLM reasoning, tool invocation, shell execution, and repository inspection.
- Architected a modular plugin framework integrating 11+ developer tools with sandboxed execution, enabling extensible and isolated tool workflows.
- Containerized the application with Docker and created GitHub Actions CI pipelines with sub-minute runtime, reducing developer onboarding from hours to under 5 minutes.

Load Balancer for 5G Networks | *C, Kubernetes, Helm, Wireshark, SCTP* **Link**

- B.Tech. thesis under Dr. Samaresh Bera; research paper (link) accepted at **IEEE INDISCON 2026**, under the Communication Systems & Network Technologies track.
- Developed a consistent-hashing load balancing algorithm that reduced average latency by 62.3% under burst traffic compared with round-robin.
- Designed an auto-scaling algorithm for AMFs to balance UE load and guide scale-out/scale-in decisions in a Kubernetes-based environment/testbed.
- Implemented stateless UE migration strategies enabling AMF de-scaling without service interruption.

Dream11 Dream Team Predictor | *FastAPI, scikit-learn, PyCaret, SHAP, Pandas, React* **Link**

- Secured **8th place among 23 IITs** at Inter IIT Tech Meet 13.0 with a predictive analytics system that generated a dream team in under 10 seconds.
- Built ETL pipelines to parse and process **18,000+** JSON match records for feature engineering and model training; achieved **29.58% MAPE** on aggregate fantasy team score prediction.
- Exposed prediction functionality through FastAPI and integrated SHAP-based explanations to make player recommendations interpretable.

Technical Skills

Languages: C/C++, Python, C#, SQL, Bash
Computer Science: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, DBMS
Backend & Web: FastAPI, REST APIs, React, TailwindCSS
Cloud & DevOps: Azure, Docker, Kubernetes, Helm, Terraform, Git, GitHub Actions, CI/CD
AI/ML: PyTorch, scikit-learn, NumPy, Pandas, OpenCV, LLM APIs

Achievements

- Selected for **Amazon ML Summer School** among the top 3,000 students, 2025.
- Secured **8th rank out of 23 IITs** at Inter IIT Tech Meet 13.0, 2024.
- Solved **600+ programming problems** across LeetCode, GeeksForGeeks, and Codeforces.
- Achieved **AIR 4036** in JEE Advanced 2022 among 155,538 students, placing in the top 2.5% nationwide.